

Unsupervised Domain-shared Convolutional Neural Network for Bearing Fault Transfer Diagnosis

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Abstract—Most previous models of intelligent fault diagnosis are built on the premise that adequate and labeled samples are taken from the same feature distribution. However, in industrial scenario, this assumption is mostly invalid. For this reason, a new method based on unsupervised domain-shared convolutional neural network is proposed for bearing fault transfer diagnosis. In this method, we propose a new method based on unsupervised domain-shared convolutional neural network (DSCNN) to extract transferable features of the original bearing vibration signals installed in two different machines. Then, the multi-kernel maximum mean discrepancy (MK-MMD) evaluation index is used for domain adaptation, trying to reduce the distribution discrepancy for source and target domains. Lastly, using the fault diagnosis knowledge of the source domain labeled samples to help classify the unlabeled samples in the target domain. We verify the effectiveness of proposed method with six transfer tasks of three bearing datasets. The results of case studies confirm that the proposed method can effectively learn the transferable features and reduce the distribution discrepancy between two domains, which also shows better transfer performance and classification accuracy compared to other methods.

Keywords—bearing, fault transfer diagnosis, domain-shared convolutional neural network, unsupervised learning

I. INTRODUCTION

Bearings are one of the most widely used components in rotating machines, different bearings faults will occur in transmission parts, which may affect the production performance and even lead to safety accidents. Intelligent fault diagnosis is an effective approach in ensuring their stable, reliable and safe operation.

Over the past few years, modern industries place a greater emphasis on system reliability and safety. This has led to a significant increase in the demand for intelligent and integrated mechanical systems. Faulty bearings can have a negative impact on the machine's performance, resulting in fatal consequences. Thus, in machinery maintenance, determining and recognizing fault locations and categories is essential [1-2].

Deep learning, as a big data processing tool, has achieved great strides in some areas like image and speech

recognition. Meanwhile, deep learning is employed to intelligently fault diagnosis, which is also receiving widespread attention. Shao *et al.* [3] presented a tracking deep wavelet auto-encoder for higher-level feature learning and automatic fault diagnosis. Wang *et al.* [4] chosen Short-Time Fourier Transform (STFT) to convert the monitoring signals into time-frequency spectrums which were served as input samples for the CNN to achieve accurate fault diagnosis. Janssens *et al.* [5] used two-dimensional CNN to identify four different types of rotating bearing states, which contained one convolution layer and connection layer, and Softmax function was used in classification layer at the top.

However, the success of these techniques is partially due to the availability of large amounts of labeled samples. In industrial scenario, firstly, the faulty samples are insufficient and unlabeled. Secondly, the complexity of manufacturing situation and variability in working conditions may cause different distribution discrepancies of these samples. For the above reasons, training an intelligent diagnostic model is challenging and how to address these challenges for fault diagnosis is urgent.

Transfer learning provides a new tool for this problem. The datasets of transfer learning task contain the source domain and the target domain, respectively. Transfer learning can reduce the number of samples needed in the target domain, and improve model performance by utilizing the knowledge from source domain. Domain adaptation, as a representative method in transfer learning, has attracted a lot of attention in recent years. It aims to explore domain-invariant structures to establish knowledge transfer from the source domain to the target domain. In rotating machines intelligent fault diagnosis, Chen *et al.* [6] proposed a fault diagnosis method based on transfer component analysis (TCA) algorithm to classify the states of bearing fault under various conditions. Han *et al.* [7] used joint distribution adaptation (JDA) for bearing fault diagnosis, which could jointly adapt marginal MMD and conditional MMD.

These methods have achieved some progress to some degree, nevertheless, when a significant distribution discrepancy exists between the two domains, such as the diagnosis cases from one machine to another machine, the generalization performance of these diagnosis methods established by shallow-level feature transfer techniques needs to be improved.

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To solve the problem, we construct a domain-shared convolutional neural network (DSCNN), which can synchronously extract the domain-invariant features of bearing signals of two different machines. Then, the evaluation index of MK-MMD is used to reduce the distribution discrepancies between the learned features. Lastly, the diagnosis knowledge obtained from labeled source domain samples is transferred to help classify the unlabeled target domain samples.

In the following parts, Section II introduces the concept of transfer learning and the basic principles of MMD. DSCNN and MK-MMD are discussed in Section III. In Section IV, six tasks are conducted for verifying the proposed method. In section V, we present our conclusions.

II. TRANSFER LEARNING AND BASIC PRINCIPLE OF MMD

A. Transfer learning

The task contains a source domain D^s and a target domain D^t . Let χ^s represents the source sample space and χ^t represents target sample space, where $\chi^s \in D^s$, $\chi^t \in D^t$, the samples x^s and x^t belongs to χ^s and χ^t , respectively. The label space $y = \{1, 2, \dots, c\}$ includes c states.

The marginal probability distribution $P(x^s)$ and $Q(x^t)$ is assumed to apply to the source samples and target samples. According to the goal of this paper, we give two definitions as follows:

(1) $D^s = \{\chi^s, P(x^s)\}$ with n^s labeled samples from $\chi^s = \{x_i^s, y_i^s\}_{i=1}^{n^s}$, which can provide diagnosis information. $D^t = \{\chi^t, P(x^t)\}$ with n^t samples from $\chi^t = \{x_i^t\}_{i=1}^{n^t}$ without any labeled samples.

(2) D^s could give sufficient bearing diagnosis knowledge for D^t , i.e., $y^t \subseteq y^s \subseteq y$, where y^s is the source domain space and y^t is the target domains space.

B. MMD

Given two datasets X , Y with n_1 and n_2 samples, respectively, where $X = \{x_i\}_{i=1}^{n_1}$ and $Y = \{y_i\}_{i=1}^{n_2}$, respectively. The formulation of MMD is given as follows:

$$D_H(x, y) := \sup_{\phi \in H} (E_{X \sim p}[\phi(x)] - E_{Y \sim q}[\phi(y)]) \quad (1)$$

where $\sup(\cdot)$ represents the supremum of the input aggregate, $\phi(\cdot)$ implies a mapping function in a reproducing kernel Hilbert space (RKHS), usually, Gaussian kernel is selected as the kernel function.

$$\langle \phi(x_i), \phi(y_i) \rangle_{H_k} = k(x_i, y_i) = \exp(-\|x_i - y_i\|^2 / \sigma) \quad (2)$$

where $k(\cdot, \cdot)$ represents the Gaussian kernel, σ is the bandwidth of kernel. Therefore, Eq. (1) can be calculated as follows:

$$\begin{aligned} D_H^2(x, y) &= \hat{D}_H^2(X, Y) = \left\| \frac{1}{n_1} \sum_{i=1}^{n_1} \phi(x_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} \phi(y_j) \right\|_H^2 \\ &= \frac{1}{n_1^2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_1} k(x_i, x_j) - \frac{2}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} k(x_i, y_j) \\ &\quad + \frac{1}{n_2^2} \sum_{i=1}^{n_2} \sum_{j=1}^{n_2} k(y_i, y_j) \end{aligned} \quad (3)$$

III. THE PROPOSED METHOD

A. DSCNN

Fig.1 shows the framework of the DSCNN. In the step of convolution operation, the shared kernel $\mathbf{k}^l \in \mathbb{R}^{H \times L \times D}$ is used to convolve to one-dimensional (1D) vector, where H , L and D is the height, length and the depth of the kernels, respectively. The output of the $(l-1)$ th layer is $\mathbf{x}_i^{l-1, D} \in \mathbb{R}^{N \times M}$, and then the output of the l th layer is given as follows.

$$\mathbf{x}_i^{l, D} = \sigma_r(\mathbf{x}_i^{l-1, D} * \mathbf{k}^l + \mathbf{b}^l) \quad (4)$$

with

$$(\mathbf{x}_i^{l-1, D} * \mathbf{k}^l)_{j, d} = \sum_{m=1}^M \sum_{h=1}^H \mathbf{x}_{(i), j+h-1, m}^{l-1, D} \cdot \mathbf{k}_{h, d}^l \quad (5)$$

where $D = \{x_i^s, x_i^t\}$ is the indexes of i th two domains, $\mathbf{b}^l$ is the bias of the first convolution layer (Conv1), $\mathbf{x}_{(i), j+h-1, m}^{l-1, D}$ is the element in the $(j+h-1)$ th row and the m th column of $\mathbf{x}_i^{l-1, D}$, and $\sigma_r(\cdot)$ means the activate function of ReLU (rectified linear unit).

Then, we choose max pooling to avoid over-fitting. The pooled features are expressed by:

$$(\mathbf{v}_i^{l, D})_{k, m} = \max\{\mathbf{x}_{(i), j, m}^{l-1, D} | j \in N^+, Z(K-1)+1 \leq j \leq ZK\} \quad (6)$$

where $\max\{\cdot\}$ returns the largest number of a vector, and Z represents the height of the non-overlapping interval.

After stacking two convolutional layers and two pooling layers, the advanced features have been flattened into a 1D vector with the first full connection layer (Full-connection1). The output of the second full connection layer (Full-connection2) is expressed as follows.

$$\mathbf{x}_i^{F_2, D} = \sigma_r(\mathbf{w}^{F_2} \mathbf{x}_i^{F_1, D} + \mathbf{b}^{F_2}) \quad (7)$$

where $\mathbf{w}^{F_2}$ is the connection weight between neurons in Full-connection1 and Full-connection2, and $\mathbf{x}_i^{F_2, D}$ is the output features of Full-connection2 with a bias $\mathbf{b}^{F_2}$.

Softmax activate function is used in the classification layer. The output of the classification layer is the probability distribution, which is presented as:

$$\mathbf{h}_i^{C, D} = \left[P(y_i^D = 1 | \mathbf{x}_i^{F_2, D}) \dots P(y_i^D = K | \mathbf{x}_i^{F_2, D}) \right]^T \quad (8)$$

$$P(y_i^D = j | \mathbf{x}_i^{F_2, D}) = \frac{\exp\left((\mathbf{w}_j^C)^T \mathbf{x}_i^{F_2, D} + \mathbf{b}^C\right)}{\sum_{j=1}^k \exp\left((\mathbf{w}_j^C)^T \mathbf{x}_i^{F_2, D} + \mathbf{b}^C\right)} \quad (9)$$

where $\mathbf{h}_i^{C, D}$ represents the probability distribution of labels for the i th sample, $\mathbf{w}_j^C$ implies the connection weight between neurons in Full-connection2 and classification layer with a bias $\mathbf{b}^C$. Table I shows details about DSCNN's parameter set.

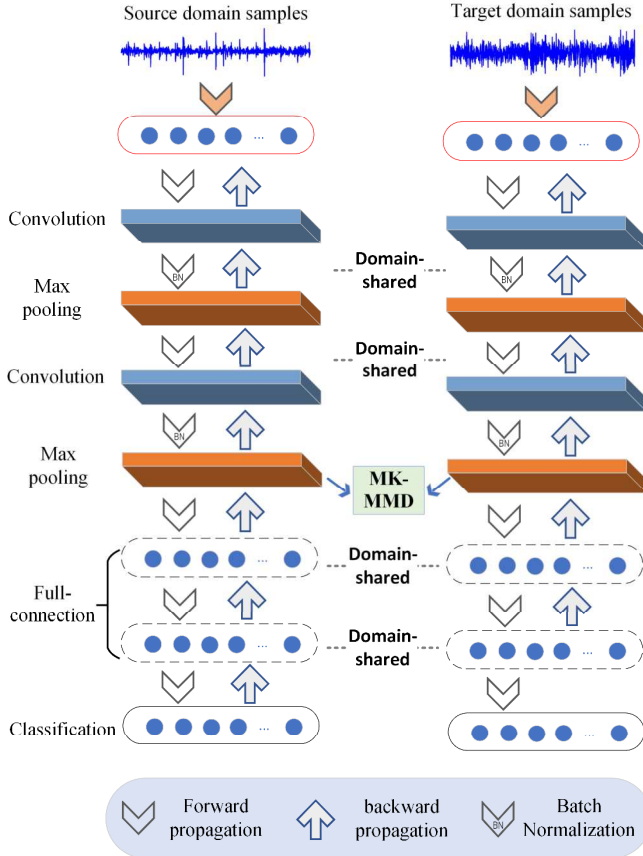


Fig. 1. The framework of DSCNN

TABLE I. PARAMETER SET OF THE PROPOSED DSCNN MODEL

Layers	Parameters	Activation functions
Input	Size: 1024×1	/
Conv1	Kernels: $15 \times 1 \times 16$	/
Batch Normalization	/	/
Max pooling	Kernels: 2, stride: 2	ReLU
Conv2	Kernels: $3 \times 1 \times 32$	/
Batch Normalization	/	/
Max pooling	Kernels: 2, stride: 2	ReLU
Full-connection1	Input size: 32×504 Output size: 8032×1	/
Full-connection2	Input size: 8032×1 Output size: 128×1	ReLU
Classification	Input size: 128×1 Output size: 4×1	Softmax

B. Domain adaptation based on Multi-kernel MMD

MMD is widely used to evaluate the distance between two distributions. Eq. (3) has given the one-kernel MMD calculation formula. However, it is important to select kernel parameters carefully to ensure optimal mapping performance. For this reason, multi-kernel MMD (MK-MMD) is employed in this paper to further improve accuracy and reduce relative error between transferable features. Generally, the convex combination $\{k_u\}$ is used to provide effective mapping estimate. The multi-kernel function is defined as follows.

$$K \square \left\{ k = \sum_{u=1}^m \beta_u k_u : \sum_{u=1}^m \beta_u = 1, \beta \geq 0, \forall u \right\} \quad (10)$$

where β_u represents the trade-off parameter of different kernels. Based on the multi-kernel function, the MK-MMD can be defined as follows

$$\begin{aligned} L_{MK-MMD}(D_s, D_t) &= \hat{D}_H^2(X, Y) = \left\| \frac{1}{n_1} \sum_{i=1}^{n_1} \phi(\mathbf{x}_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} \phi(\mathbf{y}_j) \right\|_H^2 \\ &= \frac{1}{n_1^2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_1} K(\mathbf{x}_i, \mathbf{x}_j) - \frac{2}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} K(\mathbf{x}_i, \mathbf{y}_j) \\ &\quad + \frac{1}{n_2^2} \sum_{i=1}^{n_2} \sum_{j=1}^{n_2} K(\mathbf{y}_i, \mathbf{y}_j) \end{aligned} \quad (11)$$

where $L_{MK-MMD}(D_s, D_t)$ represents the distance of MK-MMD. $K(\cdot, \cdot)$ implies the multi-kernel function.

We select Gaussian kernel as the kernel function, and the kernel number is set as 5. According to the median heuristic, the bandwidth of each Gaussian kernel is set to the median pairwise distance on the training data. Fig. 2 shows the flowchart for domain adaptation.

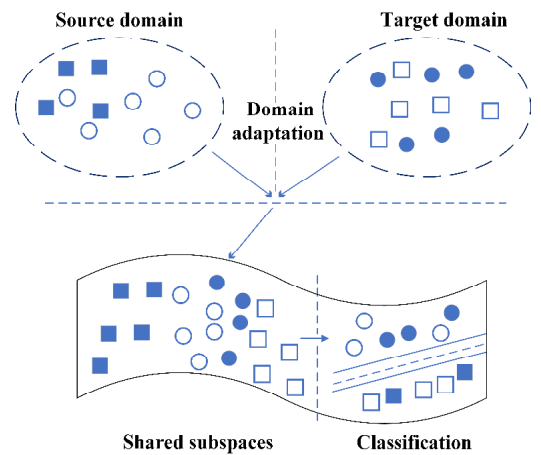


Fig. 2. Flowchart for domain adaptation.

C. Training process

Fig. 3 shows the flowchart of the DSCNN with MK-MMD, which is trained by integrating two objectives (1) Cross-entropy loss between the predicted labels of samples and their real labels; (2) the distance of MK-MMD between the transferable features of the two domains. Taking

minimization of Eq. (12) as total optimization objective, and it is defined as follows.

$$L = L_c + \lambda_{MK-MMD} L_{MK-MMD}(D_s, D_t) \quad (12)$$

$$L_c = L(\mathbf{y}_i^s, \mathbf{h}_i^{C,s}) = -(\mathbf{y}_i^s)^T \cdot \log(\mathbf{h}_i^{C,s}) \quad (13)$$

$$\lambda_{MK-MMD} = \frac{2}{1 + e^{\frac{(-10 \frac{epoch}{max-epoch})}{max-epoch}}} - 1 \quad (14)$$

where L represents the total loss function, L_c represents the cross-entropy loss function of classification layer, $\mathbf{h}_i^{C,s}$ represents the output of Softmax classification layer for the i th sample, $\mathbf{y}_i^s$ is its corresponding label, λ_{MK-MMD} implies the trade-off parameter, $epoch$ is the current training round, and $max-epoch$ means the maximum of training round.

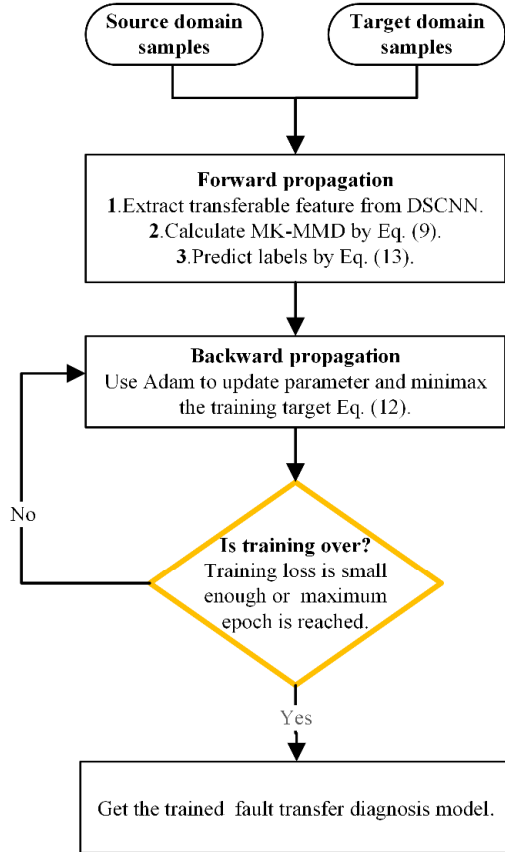


Fig. 3. Flowchart for the proposed method

D. Diagnostic process

The proposed method includes four diagnostic process in sequence:

- 1) **Domain division:** Samples in the source domain are labeled, while samples in the target domain are unlabeled.
- 2) **Feature extraction:** The source samples and the target samples are synchronously fed in DSCNN.
- 3) **Domain adaption:** Use the MK-MMD to measure the distance between two distributions, which is one of the optimization objectives during the back propagation stage.

- 4) **Fault classification:** Based on the domain adaption, classification layer is capable of classifying the unlabeled samples.

IV. CASE STUDIES

A. Dataset Description

In this paper, we select three bearing datasets (marked A, B and C) respectively collected from three different machines to verify the proposed method. Table II provides details about the datasets.

A is from Case Western Reserve University [8]. With a sampling frequency of 12 kHz, it was measured at the drive-end bearings under 0HP loads with four kinds of health states, consisting of normal (N), inner fault (IF), rolling fault (RF) and outer fault (OF). The diameter for each kind of fault was 0.07mm.

B is from roller bearings under speed fluctuation condition used in a laboratory with sampling frequency of 25.6 kHz [9]. It contains the same four kinds of health states, and the diameter for each kind of fault was 0.05mm.

C is from locomotive rolling bearings with sampling frequency of 12.8 kHz under 9800N loads with the same four kinds of health states [3].

TABLE II. INTRODUCTION TO DATASETS

Datasets	Bearing types	States	Number of samples
A	SKF6205-2RS	Normal (N)	4 × 200
B	QPZZ-II	Inner fault (IF)	4 × 200
		Rolling fault (RF)	
C	552732QT	Outer fault (OF)	4 × 200

B. Transfer diagnosis results

In this experiment, the Gaussian kernel is selected and the bandwidth of multi-kernel function is set as [0.01, 0.1, 1, 10, 100]. The batch size is chosen as 32. The initial learning rate is 0.001, whose weight decay is 1e-3.

As depicted in Fig. 4, as the epoch number increases, final MK-MMD steadily convergences during the process of distribution adaptation, indicating that the domain adaptation based on MK-MMD can reduce distribution discrepancy between two different rolling bearing datasets.

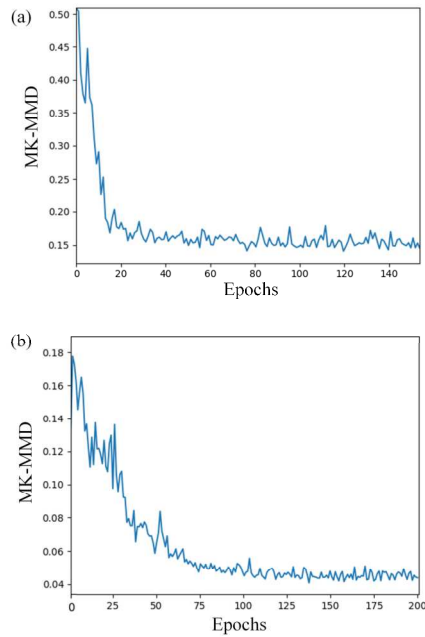


Fig. 4. Trends of MK-MMD in the epochs of transfer task (a) A→B and (b) A→C.

We compare the proposed method with three other methods to demonstrate its superiority, including CNN, TCA, and JDA. Fig. 1 and Table III shows diagnosis results for six different tasks. Each result is an average of 5 random tests.

TABLE III. CLASSIFICATION COMPARISON FOR TRANSFER TASKS.

Tasks	CNN	TCA	JDA	DSCNN+MMD (proposed)
A→B	47.50%	30.80%	50.20%	81.25%
A→C	52.15%	45.25%	53.65%	83.16%
B→C	30.75%	32.16%	40.30%	76.65%
B→A	49.35%	55.00%	52.55%	82.00%
C→A	50.75%	42.65%	50.25%	88.15%
C→B	35.65%	28.90%	45.16%	75.65%

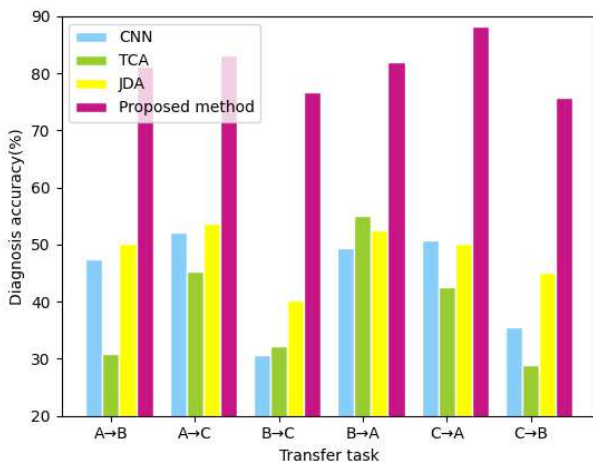


Fig. 5. Comparison of the four methods on six transfer tasks.

It is clearly to see that average classification accuracy of the proposed method is 81.43%, which is superior to the three others. The topology structure of CNN is the same as the proposed model, and the average accuracy of CNN for the diagnosis of datasets is 44.35%. Compared to the proposed method, its accuracy is smaller, which mainly because there is a tremendous difference of distribution discrepancy between the cross-machine data. The average diagnosis accuracy of TCA for six transfer tasks is 39.13% and the accuracy of JDA is 48.69%, implying that the distribution discrepancy of transferred features in two different machines has not gotten smaller. The proposed method can extract the deep-layer features directly from raw acceleration data, which can greatly improve cross-machine bearing diagnosis accuracy.

C. Visualization of the transferable features

To visualize the effectiveness of MK-MMD, the t-SNE [10] is employed in Conv2. The results of CNN and the proposed method after t-SNE are plotted in Fig. 6.

The proposed method broadly divides the data into four clusters, moreover, the same category of two domains is almost in one cluster, while the cluster boundary formed by CNN is not clear and cannot be divided into four clusters, and the two domains have not fully mixed into one cluster.

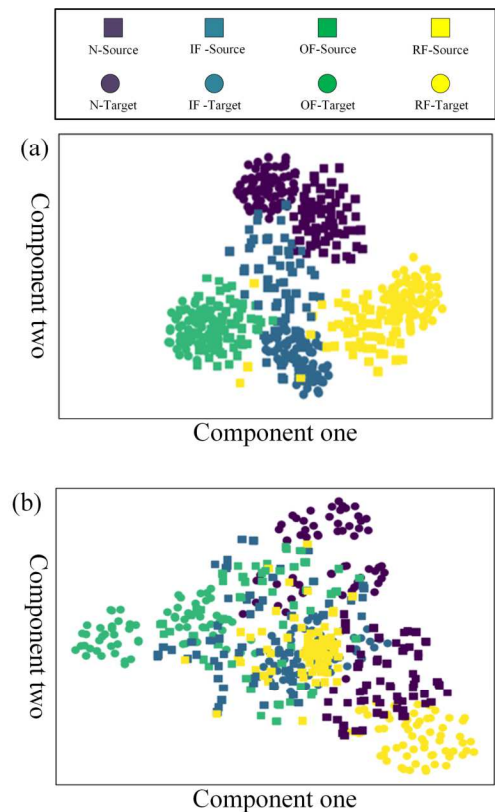


Fig. 6. 2D visualizations of the transferable features in transfer task B→A: (a) the proposed method and (b) CNN.

V. CONCLUSION

Aiming to achieve cross-machine fault diagnosis of bearing, this paper presents DSCNN to synchronously extract transferable features of bearing vibration signals. Then, MK-MMD is used for domain adaptation, trying to reduce the distribution discrepancy of transferable features.

Lastly, diagnosis knowledge can be transferred from the labeled source domain to the unlabeled target domain.

The proposed method is verified with six transfer tasks. The results illustrate that the proposed method could learn transferable features in different machines and perform higher compared with other methods. The future work is to study methods such as multi-layer MMD and under-sampling to solve more complex fault diagnosis tasks.

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